



SmartWaste AI Manager

Turning Waste Reports into Immediate Action



Snap a Photo

Citizen captures a waste image on-site



AI Verifies It

Gemini AI detects type, severity & risk



Authorities Take Action

Instant alert + logged report, automatically



The Problem

Waste reporting today is slow, vague, and easy to ignore.



Reported Too Late

Waste complaints reach authorities long after piles have grown — delaying cleanup and worsening public health risk.



No Verified Evidence

Authorities lack accurate location data and visual proof, making it hard to prioritize or dispatch teams.



Manual, Slow Process

Manual registers and helpline calls delay emergency response and increase sanitation & health risks.

Existing alternatives fall short:

Municipal helpline calls

Government complaint apps

Email-based complaints

Manual complaint registers

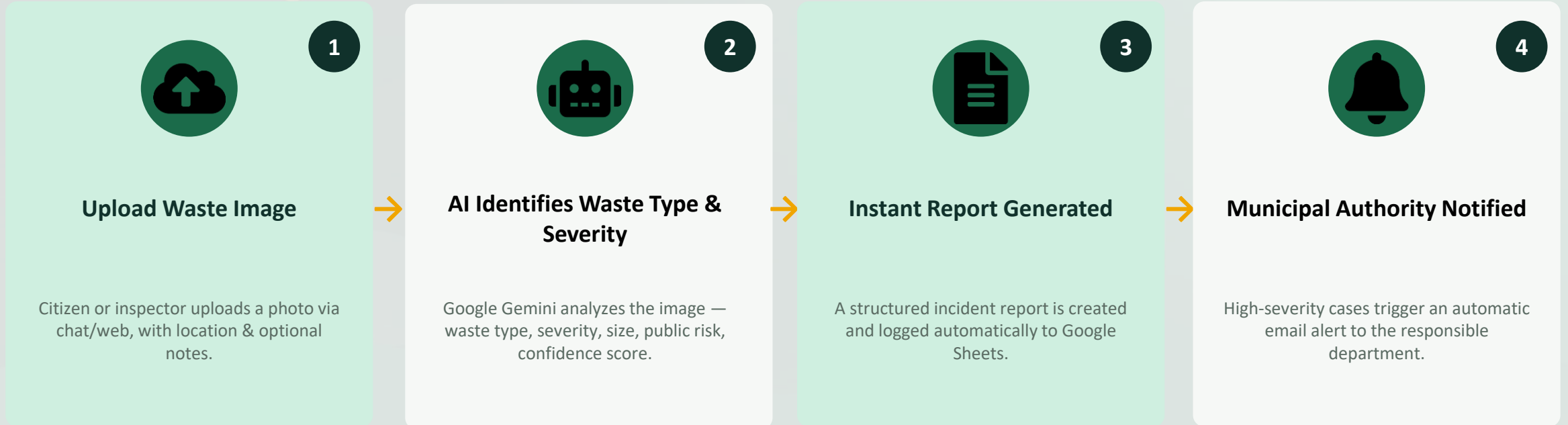
Physical waste inspections

Citizens need a fast, reliable way to report waste — municipalities need verified, AI-powered reports for quicker action.



Our Solution

An AI agent that turns a single photo into a verified, actioned incident report.



Result: what used to take days of manual follow-up now happens in seconds — fully automated, end to end.

Why We're Different

Unique Value Proposition

"AI transforms a simple waste photo into a verified incident report — with severity analysis, evidence, and instant municipal alerts."

-  Verifies waste using AI — not just collecting complaints
-  Prioritizes severity automatically (Low → Critical)
-  Stores visual evidence for accountability
-  Auto-alerts the responsible authority — no manual routing



Unfair Advantage

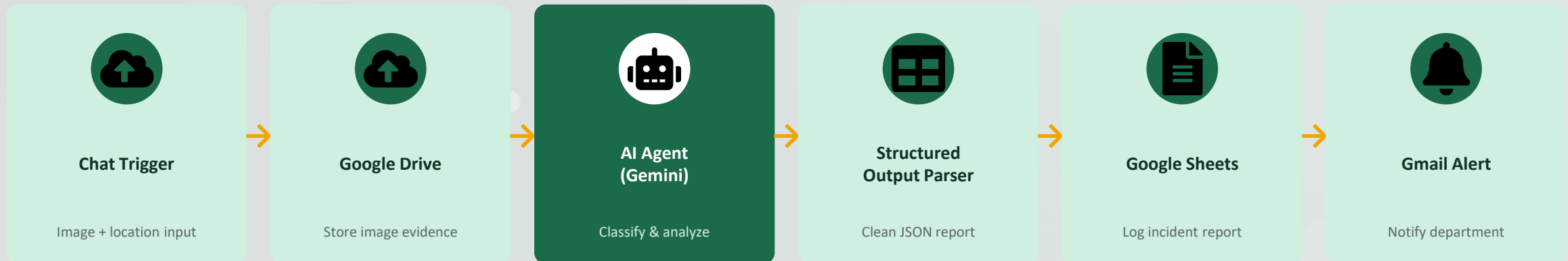
An end-to-end AI-driven reporting workflow that transforms a single waste image into a verified incident report — complete with evidence, prioritization, and automated authority notification.

***Zero manual routing. Zero missed reports.
From photo to municipal alert — fully automated in one workflow.***



System Architecture

Built as an automated AI agent workflow in n8n — no manual glue code.



Conditional logic (IF node): only High / Critical severity reports auto-trigger a Gmail alert to the municipal department — keeping inboxes noise-free.

Tech: n8n (orchestration) · Google Gemini (vision + reasoning) · Google Drive/Sheets/Gmail (storage & alerts)

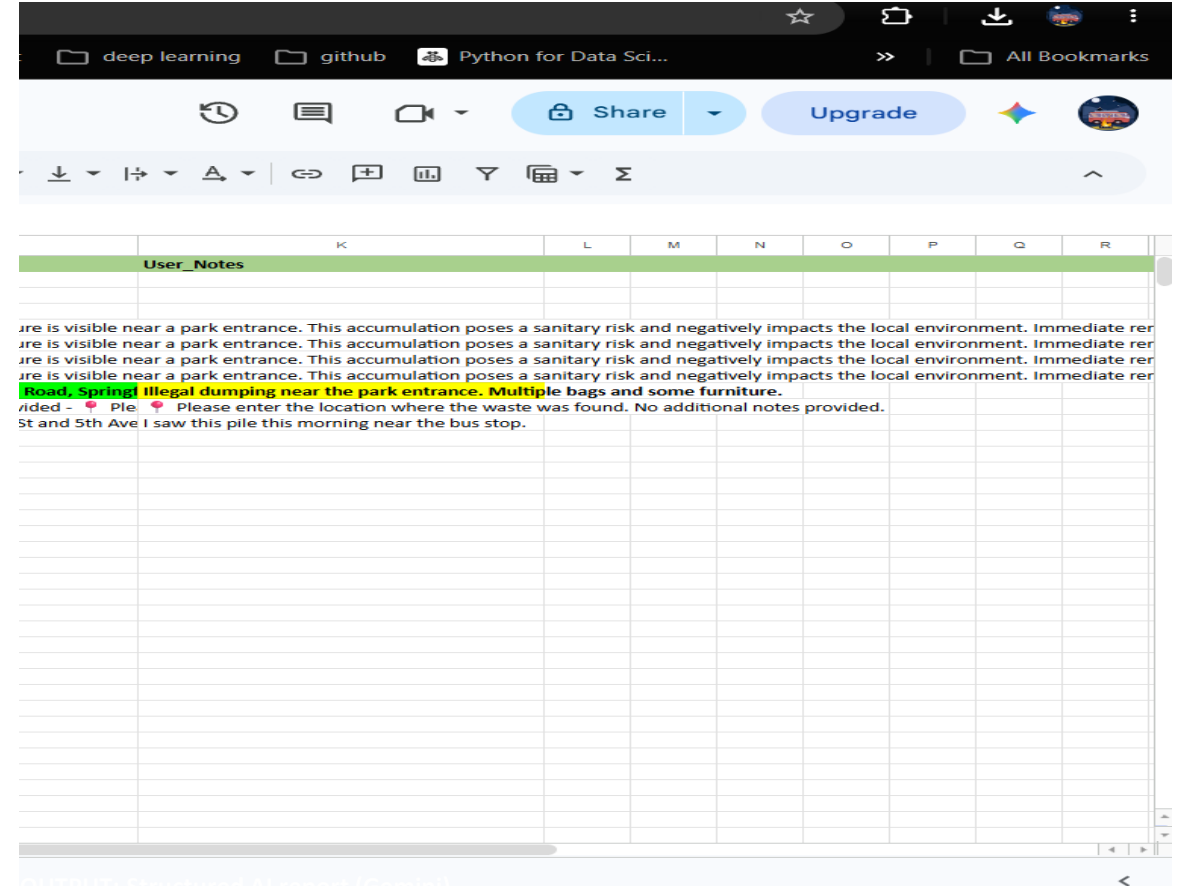


A single roadside photo goes straight into the AI agent for classification.



INPUT: Uploaded roadside waste image

Location: Dwarka Road · Note: "please clear it fast, it's causing pollution"





Live Demo — From Report to Action

Every report is logged and, if severe, sent straight to the responsible department.

The screenshot shows a Gmail interface. On the left is a sidebar with folders: Compose, Inbox (1,258), Starred, Snoozed, Sent, Drafts (3), Purchases (24), and More. The main area displays an email from Aryan Sable (sablearyan128@gmail.com) with the subject 'SmartWaste AI Alert – Immediate Action Required'. The email body reads: 'Hello Municipal Team, A new waste incident has been detected by SmartWaste AI Manager. 🗑️ Waste Type: Mixed Waste 🚨 Severity: High ⚠️ Public Risk: Medium 🏢 Department: Municipal Waste Management Department 📋 Recommended Action: Deploy a waste collection team for immediate removal and thorough cleaning of the area. 📄 Summary: A large accumulation of mixed waste, including cardboard, plastic, and general debris, is visible on a public sidewalk corner. This pile obstructs pedestrian traffic and poses sanitation risks, requiring urgent removal to maintain public health and safety. The sanitation department should prioritize this site for immediate cleanup. This alert was generated automatically by SmartWaste AI Manager.'



Auto-logged to Google Sheets — every incident, timestamped & tracked

The screenshot shows a Google Drive interface. The left sidebar lists navigation options: New, Home, Projects, My Drive, Computers, Shared with me, Recent, Starred, Spam, Bin, and Storage (1.46 GB of 15 GB used). The main area shows the 'Smart_Waste AI Reports' folder. At the top, there's a bar with 'Ask Gemini' and the text 'Summarise, analyse and get up to speed with files in this folder.' Below this is a table of files:

Name	Owner	Date modified	File size	Sort
🗑️ Waste_20260705215403.jpg	me	21:56 me	167 KB	⋮
🗑️ Waste_20260705220617.jpg	me	22:06 me	3.2 MB	⋮



Tech Stack

A low-code, API-driven stack — fast to build, easy to scale.



n8n

Workflow orchestration — trigger, logic, routing & conditional alerts



Google Gemini

Vision + reasoning model — classifies waste type, severity, risk



Structured Output Parser

Enforces clean, consistent JSON schema for every report



Google Drive

Stores uploaded image evidence with shareable links



Google Sheets

Acts as the live incident database / report log



Gmail API

Sends automated high-severity alerts to departments

Who It's For

Primary Customers



Municipal Corporations



Smart City Authorities



Housing Societies



Universities & Campuses

Secondary Customers



NGOs



Citizens



Corporate Campuses

Distribution channels: mobile/web app, municipal corporation portals, smart city partnerships, NGO & community programs, automated email notifications.



Business Model

Cost Structure

Fixed Costs

- Cloud hosting & AI APIs
- Development & maintenance
- Security & database infrastructure

Variable Costs

- Image processing
- Email notifications & cloud storage
- API usage based on report volume

Revenue Streams

Revenue Model

Annual SaaS subscription for municipalities; enterprise licensing for campuses & organizations

Lifetime Value

Long-term government contracts + monthly subscription & deployment fees

Gross Margin

High-margin cloud-based AI platform — costs scale sub-linearly with usage

Key Metrics & Success

How we measure progress and impact.



Images Analyzed



Reports Submitted



High-Severity Incidents Detected



Alerts Delivered Successfully



Avg. Report-to-Alert Time

Success Criteria

-  Faster municipal response times
-  Higher citizen participation in reporting
-  Increased reporting accuracy via AI verification
-  Reduction in unattended waste cases



Impact & What's Next

Feasible today. Built to scale across cities.

Long-Term Impact



Cleaner public spaces via faster, verified cleanup response



Reduced health & sanitation risks in high-density areas



Data-driven waste management for smart cities (SDG 13)



Empowers citizens to be active partners in civic action

Future Roadmap

1

Native mobile app for one-tap citizen reporting

2

GPS auto-tagging & live municipal dashboard

3

Analytics on waste hotspots & recurring dump zones

4

Multi-language support for wider adoption

Together, Let's Build Cleaner and Smarter Cities with AI
Thank You!— Questions?